UNIT – II

MANUFACTURING PROCESS

Manufacturing is an activity of turning raw Material into finished Products to be used for some Purpose. It also involves assembly of Parts to make final Product.

Types of Manufacturing Processes

1) Primary manufacturing Processes: It involves the Conversion of initial Material into Semi-finished Products

Ex: Casting, rolling, forging, and extrusion. etc.

2) **Secondary Manufacturing Processes:** In this Process the accurate size of final Products are obtained.

Ex: Coining, drawing, sheet-forming etc.

Based on the Primary and Secondary Manufacturing process We Can classified as

(a) Casting [Starting Material is liquid)

(b) Metal -forming (or) Metal Working Pocesses [Starting Material is "Solid which consists of Plasticity.

(c) Machining (or) cutting: Starting Material is solid which extra material is removed.

(d) Powder metallurgy (or) Particulate process: starting material is powder which is compacted and sintered.

(e) Joining: In this process two or more components are joined together to produce the required product. Examples are welding, brazing, soldering etc.

(f) Assemble processes: the assemble processes for machines and mechanisms is the part of the manufacturing process concerned with the different joining of the finished parts.

(g) Surface treatment process: Unwanted material like oil, dirt, will, removed by this process.

Selection of manufacturing Process:

Many factors are involved in Selection of Manufacturing Process. Some of the considerations are.

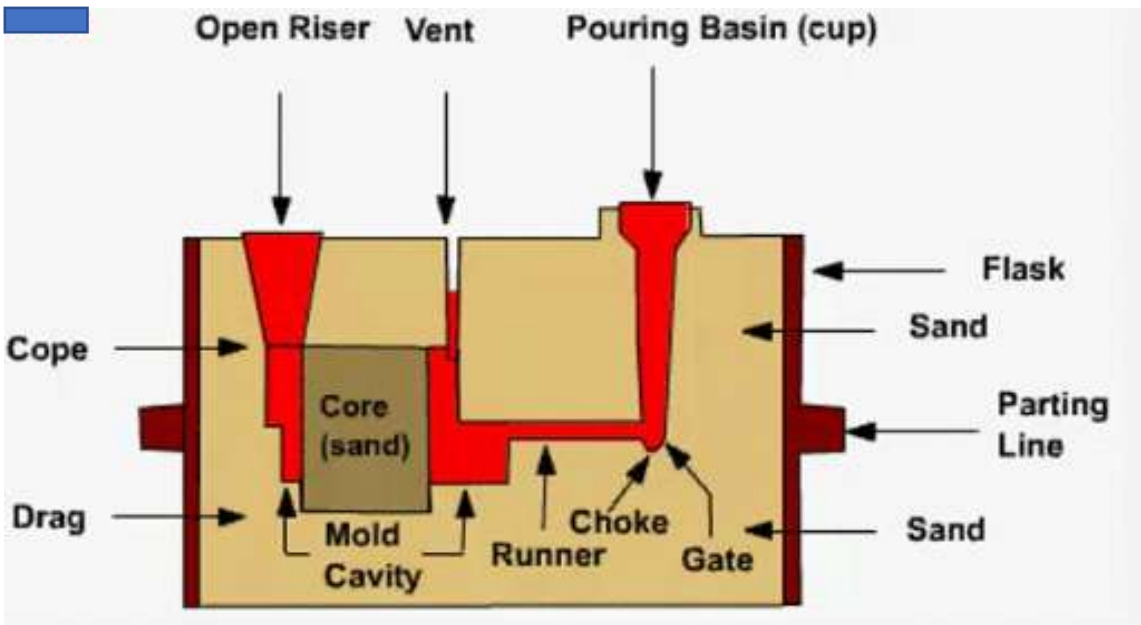
1) Material to be used
2) Shape and size of the Components
3) Accuracy and Surface -finish
4) volume of Production
5) Economy.

Importance

Now-a-days the economic development of Country is completely based on Material resources, skill and energies expand in utilising resources. ie. Manufacturing. Manufacturing is a Series of complex interaction between Materials, Machines, energy and People. It begins with Certain of individual Parts that will finally assembled to Produce final Products

Casting

Casting is defined as “the shaping of Material in liquid State”. Here the liquid Metal is poured in a mould Cavity, usually made of Sand, and it is allowed to Solidity after completion of Solidification the fart is removed from mould either by breaking Core, Separating the two Parts of mould. The part obtained by casting is called "Casting (or) founding.



Drag: Lower-Moulding flask

Cope: Upper - Moulding flask

Pattern: It is replica of a final object-to be made. The mould cavity is made with help of Pattern.

Parting line: It is a dividing line b/w 2 Moulding flask.

Facing Sand: It is a small amount of Carbonaceous Material Sprinkled on the inner Surface of Moulding cavity to Provide better Surface finish.

Moulding Sand: It is a mixture of silica, clay, moisture with Certain Proportions.

Care: It is used for making hollow Cavities in Castings.

Pouring Basin: It is a Small-shaped Cavity at the top of the mould into which molten metal is Poured

Sprue: It is a Passage through the molten metal -from the Pouring basin.

Runner. It is a Passage ways, which molten metal flow is regulated before they reach the mould Cavity..

Gate: The actual entry Point through molten metal enters the mould cavity.

Riser: It is a reservoir of molten metal Provided in the Casting, So that hot metal Can flow back into the mould cavity.

<u>Steps involved in Casting:</u>

Steps involved in casting: Now-a-days 75% of the total Outlet of castings are made in Sand Moulding. So the steps involved in Saud Moulding are

1) Selection of mould box/ flask:

of Casting Based on the requirement of casting we can Select the mould, which must be of Proper size to adjust mould: Cavity, riser, and gating System etc.

2) Preparation of Drag:

- Initially the drag will be placed on bottom boar
- "Facing Sand" is Sprinkled Care-fully around the drag Pattern' So that the Pattern does not Stick with moulding Sand during removal of Pattern
- The drage is filled with moulding Sand and rammed this sand around the Pattern, the ramming Process is close 3(or) 4 times.
- The excess amount of Sand is removed with help of Strike off bar.

- The drag is the rolled over by 180 and the Parting Sand is Sprinkled over on the top of stray.

3) Preparation of cope:

- The cope Pattern is placed on the help of dowel Pins.
- The pattern Sand is sprinkled carefully around the Cope Pattern.
- Runner Pin [Sprue] and Riser Pins are located in Vertical Position at Suitable locations. These are help to form Suitable Sizes of cavities for Pouring molten metal.
- Now the cope will be filled with moulding Sand and ramming Process is done. and excess Material is removed
- Remove sprue and riser Pins and Create vent holes. These are helps to "escape of gases during Pouring and Solidification of casting.

4) Cutting of Gate & Pouring of Metal:

- Remove both the Cope and drag Patterns, and repair the mould Suitable if needed and dressing is applied.
- The gate is then cut connecting the lower base of sprue basin with runner and then the mould Cavity.
- The Cope is then clamped with drag and the mould ily ready for Pouring

Advantages:

1. Design flexibility
2. Reduced costs
3. Dimensional accuracy
4. Versatility in production

Disadvantages:

1. Lot of molten metal is wasted in riser & gating.
2. Casting may require machining to remove rough surfaces.

Applications:

Automobile parts (engine block, cylinder head)
Machine components (machine beds, pumps, valves)
Pipes and pipe fittings
Turbine blades (aerospace)
Railway wheels and parts
Manhole covers (construction)

FORMING

"Forming is a manufacturing process in which the shape of a solid metal is changed by plastic deformation under applied force, without removing or melting the material."

Types of Forming Process:

1) **Bulk forming:**
   Bulk forming is a forming process in which a large amount of metal is plastically deformed, usually under compressive forces, to change its shape.

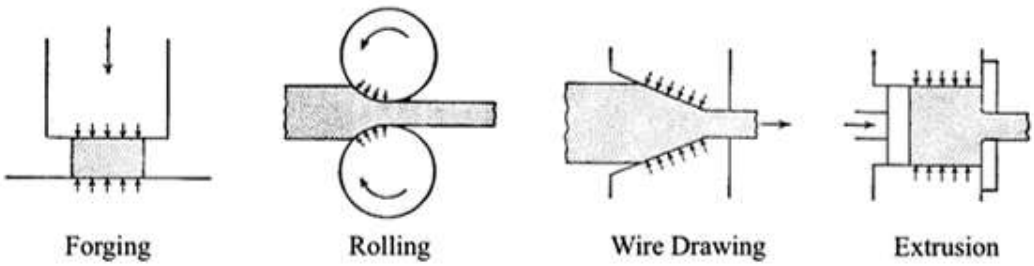
Examples:

Rolling: In this process, the work piece in the form of slab or plate is compressed between two rotating rolls in the thickness direction, so that the thickness is reduced. The rotating roll draws the slab into the gap and compresses it. The final product is in the form of sheet.

Forging: The work piece is compressed between two dies containing shaped contours. The die shapes are imparted into the final part.

Extrusion: In this, the work piece is compressed or pushed into the die opening to take the shape of the die hole as its cross section.

Wire drawing: similar to extrusion, except that the work piece is pulled through the die opening to take the cross-section.



2) **Sheet metal Forming:**

Sheet metal forming is a process in which thin metal sheets are plastically deformed to obtain the required shape without removing material.

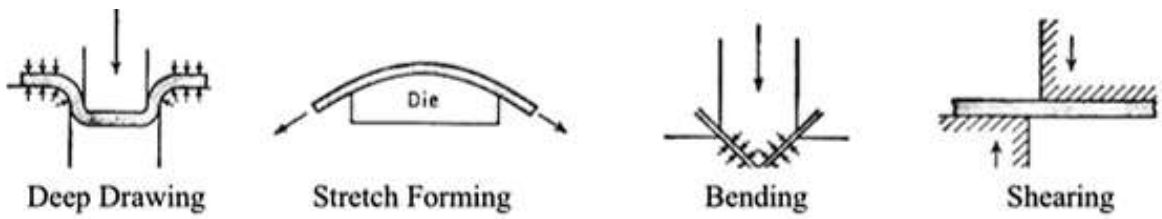
Examples:

Sheet forming: Sheet metal forming involves forming and cutting operations performed on metal sheets, strips, and coils. The surface area-to-volume ratio of the starting metal is relatively high. Tools include punch, die that are used to deform the sheets.

Bending: In this, the sheet material is strained by punch to give a bend shape (angle shape) usually in a straight axis.

Deep (or cup) drawing: In this operation, forming of a flat metal sheet into a hollow or concave shape like a cup, is performed by stretching the metal in some regions. A blank holder is used to clamp the blank on the die, while the punch pushes into the sheet metal. The sheet is drawn into the die hole taking the shape of the cavity.

Shearing: This is nothing but cutting of sheets by shearing action.



Advantages:

Good surface finish
High production rate (mass production possible)
Less material waste
High dimensional accuracy
Suitable for complex shapes (thin components)
Lightweight products

Disadvantages:

Suitable only for thin sheets
High initial tooling cost (dies, presses)
Chances of defects (wrinkling, tearing)
Limited thickness range
Skilled operation required

Applications:

Automobile body parts (doors, panels)
Aircraft skin and structural parts
Kitchen utensils (cups, containers)
Electrical enclosures (switch boxes)
Metal cans and packaging
Roofing sheets and panels

Joining

Joining is one of the manufacturing processes that are used to assemble two or more solid components or structures together. It can be employed to permanently or temporarily join any type of engineering materials such as metals, ceramics, plastics and composites. In order to cater the need to join wide variety of materials in several ways, there exist various joining processes, which can be broadly classified into two groups

Temporary joining: As the name suggests, a temporary joint is not fixed. That means it allows easy and quick dismantling of jointed structures without breaking them. This is very Page useful for inspection, maintenance and repairing purposes. However, such joint may not necessarily be strong and leak-proof. Typical joining processes that allow temporary joints include fastening, cotter joint, knuckle joint, key joint, press fit, seam joint, etc.

2. Permanent joining: A permanent joint is meant to stay fixed for long duration. It does not allow disassembly of jointed structures without breaking them. It possesses challenge during inspection, maintenance and repairing purposes as sometimes component or structure parts are required to damage. However, it offers a sound, reliable and leak-proof joint with high load carrying capacity. Typical permanent joining processes include welding, riveting, soldering, brazing, coupling, adhesive joining, etc.

Advantages

1. Complex shapes possible
2. Large structures can be made
3. Different materials can be joined
4. Repair work easy

Disadvantages

1. Joint may be weak compared to base metal
2. Defects may occur
3. Inspection required
4. Some processes are costly

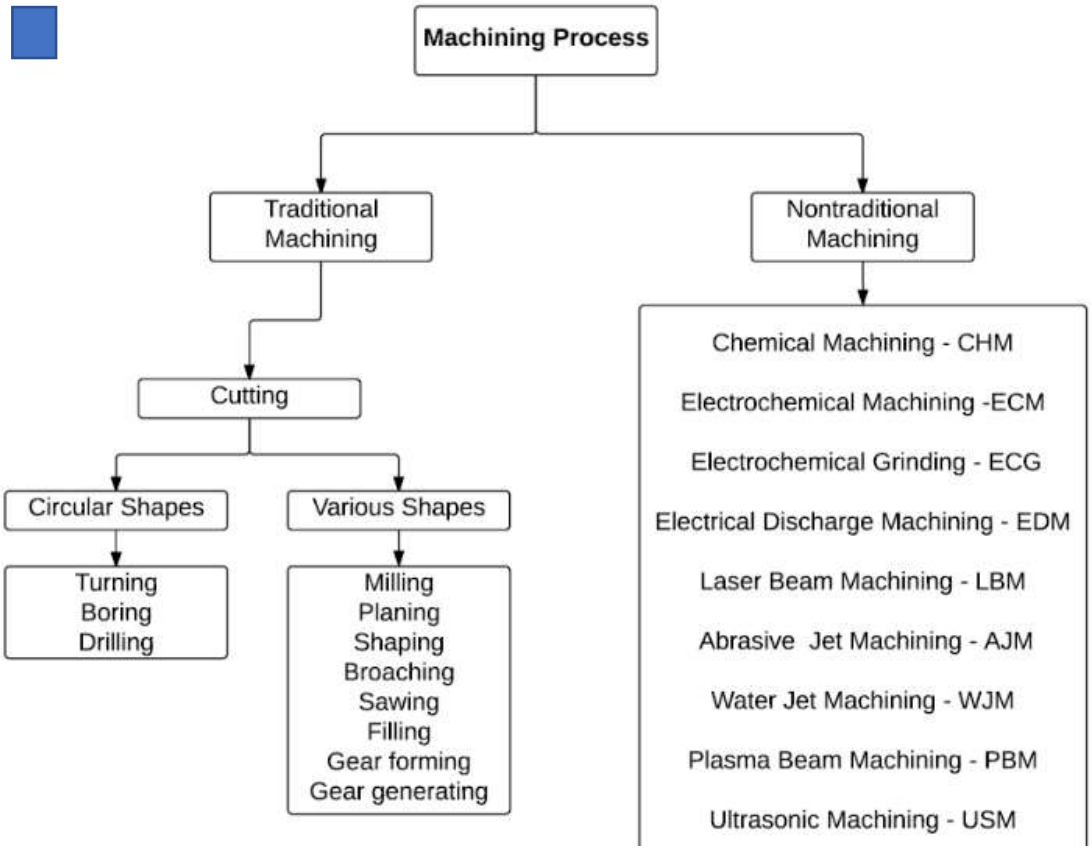
Applications

1. Automobile (car body, chassis)
2. Construction (bridges, buildings)
3. Aerospace (aircraft structures)
4. Electronics (circuit boards)
5. Shipbuilding

Machining

Machining is a manufacturing process in which material is removed from a workpiece in the form of chips using cutting tools to obtain the required shape, size, and surface finish.

Classification of Machining Process:



Traditional Machining process:

Traditional machining is a process where sharp cutting tools physically cut the material to get the required shape and size.

Turning: Turning is a machining process in which a rotating workpiece is cut by a single-point cutting tool to produce cylindrical shapes.

Boring: Boring is a machining process used to enlarge and finish an already existing hole using a single-point cutting tool**.**

Drilling: Drilling is a machining process used to create a circular hole in a workpiece using a rotating drill bit (multi-point tool)**.**

Milling: Milling is a machining process in which material is removed from a workpiece using a rotating multi-point cutting tool to produce flat or complex surfaces.

Planing: Planing is a machining process in which the workpiece moves back and forth**,** and the cutting tool remains stationary to produce flat surfaces.

Shaping: Shaping is a machining process in which the cutting tool moves back and forth, while the workpiece remains stationary to produce flat surfaces.

Broaching: Broaching is a machining process in which material is removed using a multi-point cutting tool called a broach, where teeth are arranged in a series to remove material in one pass.

Sawing: Sawing is a machining process used to cut a workpiece into two or more parts using a toothed blade.

Filing: Filing is a machining/finishing process in which small amounts of material are removed using a file (multi-point cutting tool) to obtain smooth surface, correct size, and shape.

Gear generation: Gear generating is a machining process in which gear teeth are produced by a relative rolling motion between tool and workpiece, following the correct tooth profile.

Non traditional machining Process:

Non-traditional machining is a process in which material is removed without direct contact between tool and workpiece, using electrical, thermal, chemical, or high-energy methods**.**

Advantages:

High accuracy and precision
Good surface finish
Suitable for complex shapes
Can machine hard materials
Tight tolerances can be achieved
Good dimensional control
Suitable for small batch production

Disadvantages:

High material wastage (chips)
Time-consuming process
High labor and machine cost
Tool wear occurs
Requires skilled operators
Not economical for mass production
Energy consumption is high

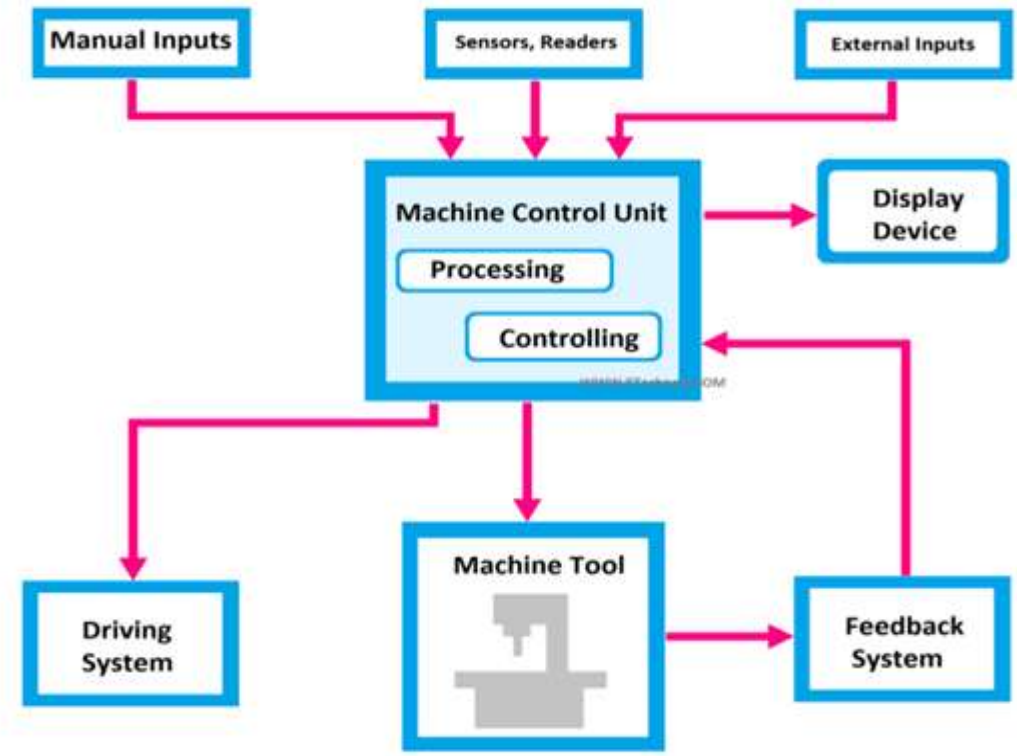
Applications:

Manufacturing machine parts (shafts, gears, bolts)
Automobile components
Aerospace components
Tool and die making
Repair and maintenance works
Precision components manufacturing

CNC MACHINES

Introduction to CNC machines: CNC means Computer Numerical Control. When computers are used to control a Numerical Control (NC) machine tool than the machine is called CNC machine. In other words the use of computers to control machine tools like lathe, mills, shaper etc is called CNC machine. The cutting operations performed by the CNC is called CNC machining, in CNC machining, programs are designed or prepared first and then it is fed to the CNC

machine. According to the program, the CNC controls the motion and speed of the machine tools.



The main parts of the CNC machine are:

1) Input Devices: These are the devices which are used to input the part program in the CNC machine. There are three commonly used input devices and these are punching tape reader, magnetic tape reader and computer via RS-232-C communication.

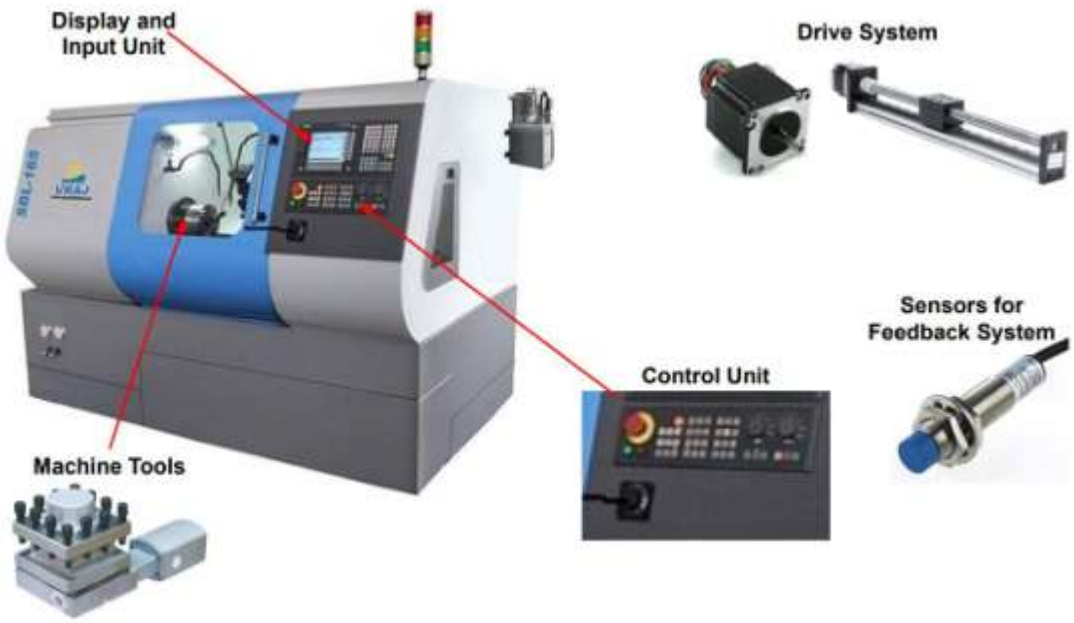
2) Machine Control Unit (MCU): It is the heart of the CNC machine. It performs all the controlling action of the CNC machine, the various functions performed by the MCU are a. It reads the coded instructions fed into it. b. It decodes the coded instruction

3) Machine Tool: A CNC machine tool always has a slide table and a spindle to control of the position and speed. The machine table is controlled in X and Y axis direction and the spindle is controlled in the Z axis direction.

4) Driving System: The driving system of a CNC machine consists of amplifier circuits, drive motors and ball lead screw. The MCU feeds the signals (i.e. of position and speed) of each axis to the amplifier circuits. And the actuated drive motors rotate the ball lead screw to position the machine table.

5) Feedback System: This system consists of transducers that acts like sensors. It is also called as measuring system. It contains position and speed transducers that continuously monitor the position and speed of the cutting tool located at any instant. The MCU receives the signals from these transducers and it uses the difference between the reference signals and feedback signals to generate the control signals for correcting the position and speed errors.

6) **Display Unit:** A monitor is used to display the programs, commands and other useful data of CNC machine



Different Types of CNC Machines:
There are several types of CNC machines
1) Mill CNC Machine
2) Lathe CNC Machine
3) Drilling CNC Machine
4) Grinding CNC Machine
5) Laser Cutting CNC Machine
6) Plasma Cutting CNC Machine
7) Water jet Cutting CNC Machine
8) EDM (Electric Discharge Machine)

Advantages:
1) It can produce jobs with the highest accuracy and precision than any other manual machine.
2) It can be run for 24 hours a day.
3) The parts produced by it have the same accuracy. There is no variation in the parts manufactured by a CNC machine.
4) A highly skilled operator is not required to operate a CNC machine. A semi-skilled operator can also operate accurately and more precisely.
5) Operators can easily make changes and improvements and reduces the delay time.
6) It has the capability to produce complex designs with high accuracy in minimum possible time.
7) The modern design software, allows the designer to simulate the manufacturer of his/her idea. And this removes the need for making a prototype or model and saves time and money.
8) Fewer workers are required to operate a CNC machine and save labor cost.

Disadvantages:
1) The cost of the CNC machine is very high as compared with a manually operated machine.
2) The parts of the CNC machines are expensive.
3) The maintenance cost in the case of CNC is quite high.
4) It does not eliminate the need for costly tools.

Application:
Almost every manufacturing industry uses CNC machines. With an increase in the competitive environment and demands, the demand for CNC usage has increased to a greater extent. The machine tools that come with the CNC are lathe, mills, shaper, welding, etc. The industries that are using CNC machines are the automotive industry, metal removing industries, industries of fabricating metals, electrical discharge machining industries, wood industries, etc.

3D printing:

3D printing is the process of creating a three dimensional object from a digital design by building up layers of material with a 3D printer. 3D printing is also known as additive manufacturing because material is added during the process of manufacturing to form the object. The opposite of this, known as subtractive manufacturing, is where material is cut away from a sheet or a block of material to form the object, for example manufacturing processes such as milling or laser cutting.

Where is 3D printing used: 3D printing is now being used across a wide range of industries including engineering, automotive, aerospace, construction, architecture, medicine, product design, food and fashion.

Examples:

a. Used for rapid prototyping of parts and components in engineering, aerospace and automotive industry.
b. Creation of custom jigs and templates for the automotive industry.
c. In medicine, 3D printing from patient scans to assist in planning operations or for student training.
d. Production of custom low cost prosthetics and disability aids.
e. 3D printing shelters and humanitarian aids.
f. Small enterprises creating personalised 3D printed products and gifts.
g. 3D printing shoes,

Overview of the 3D printing process:

The 3D printing process consists of a number of stages from initial design to production of a 3D printed object.

1) **3D modeling:** The 3D printing process starts with a 3D model. You can design your own 3D model using 3D CAD software such as TinkerCAD, you can also download free models that have been designed by others from 3D model repositories such as these can be edited in 3D CAD software if required to customise the designs to your own requirements.

2) **Saving the 3D model:** When the 3D model is complete, it needs to be saved as a suitable format. The most common 3D -le type is .STL so save the -le as a .STL -le where possible. The Ultimaker software will also support the .OBJ .3MF formats, so if you cannot save as a .STL save as one of these.

3) **Preparing the 3D model for printing:** Before you can 3D print a CAD model, the model needs to be sliced and code needs to be generated that the machine can understand. This is done in special 3D printer slicing software. The Ultilmaker slicing software is called Cura.

Advantages:

Complex shapes can be easily created
Less material waste
Fast prototyping
Easy customization
No need for tooling
Lightweight parts can be made
Suitable for small production

Disadvantages:

Slow production speed
Limited materials available
High machine cost
Poor surface finish (needs post-processing)
Lower strength in some cases
Size limitation
Skilled operators required

Applications:

Medical field (prosthetics, implants)
Automobile industry (prototype parts)
Aerospace (lightweight components)
Education (models)
Architecture (building models)
Consumer products (toys, jewelry)
Tooling (jigs and fixtures)